



BITS Pilani
K K Birla Goa Campus

8086/8088 Pins

MICROPROCESSORS & INTERFACING (CS F241)

Dr. Manideepa Mukherjee
Department of CS & IS



8086/8088 Pin Out

8086 CPU										MIN MODE (MAX MODE)									
GND	1	←								40	VCC								
AD14	2	↔								39	AD15	↔							
AD13	3	↔								38	A16/S3	←							
AD12	4	↔								37	A17/S4	←							
AD11	5	↔								36	A18/S5	←							
AD10	6	↔								35	A19/S6	←							
AD9	7	↔								34	BHE/S7	←							
AD8	8	↔								33	MN/MX	←							
AD7	9	↔								32	RD	←							
AD6	10	↔								31	Hold	←							
AD5	11	↔								30	HLD/A	←							
AD4	12	↔								29	WR	←							
AD3	13	↔								28	M/IO	←							
AD2	14	↔								27	DT/R	←							
AD1	15	↔								26	DEN	←							
AD0	16	↔								25	ALE	←							
NMI	17	→								24	INTA	←							
INTR	18	→								23	TEST	←							
CLK	19	→								22	READY	←							
GND	20	←								21	RESET	←							

MN/MX (PIN 33)

The 8086 microprocessor can work in two modes of operations: Minimum mode and Maximum mode.

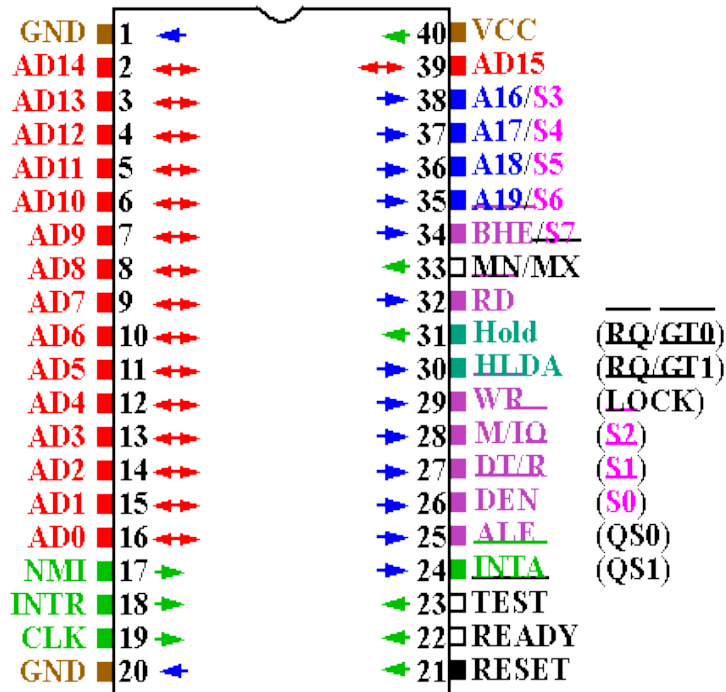
In the **minimum mode** ($MN/\overline{MX} = \text{High}$) microprocessor do not associate with co-processors mode of operation and can not be used for multiprocessor systems.

In the **maximum mode** ($MN/\overline{MX} = \text{Low}$) 8086 can be used in multiprocessor or co-processor configuration

8086/8088 Pin Out

8086 CPU

MIN MODE (MAX MODE)



In Minimum mode pin 33 is tied to VCC. 8086 itself generates all the bus control signals.

DT/R' (Data Transmit/Receive) Output signal from the processor to control the **direction of data** flow through the data transceivers. If **1 transmit if 0 receive**.

DEN (Data Enable) Output signal from the processor used as **to activate the external data bus buffer**.

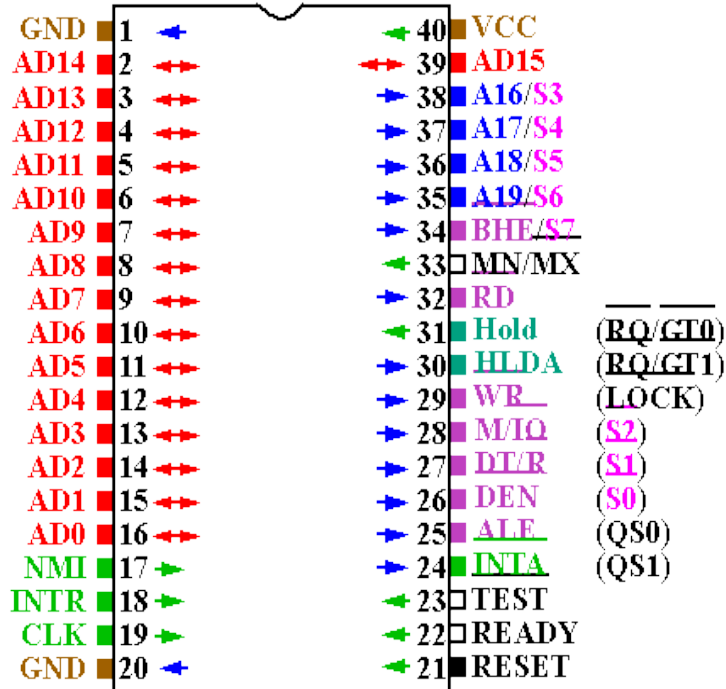
ALE (Address Latch Enable) Used to **demultiplex the address and data lines** using external latches

M/I $\bar{O}$: Used to differentiate **memory access and I/O access**. For memory reference instructions, it is high. For IN and OUT instructions, it is low.

8086/8088 Pin Out

8086 CPU

MIN MODE (MAX MODE)



In Minimum mode pin 33 is tied to VCC. 8086 itself generates all the bus control signals.

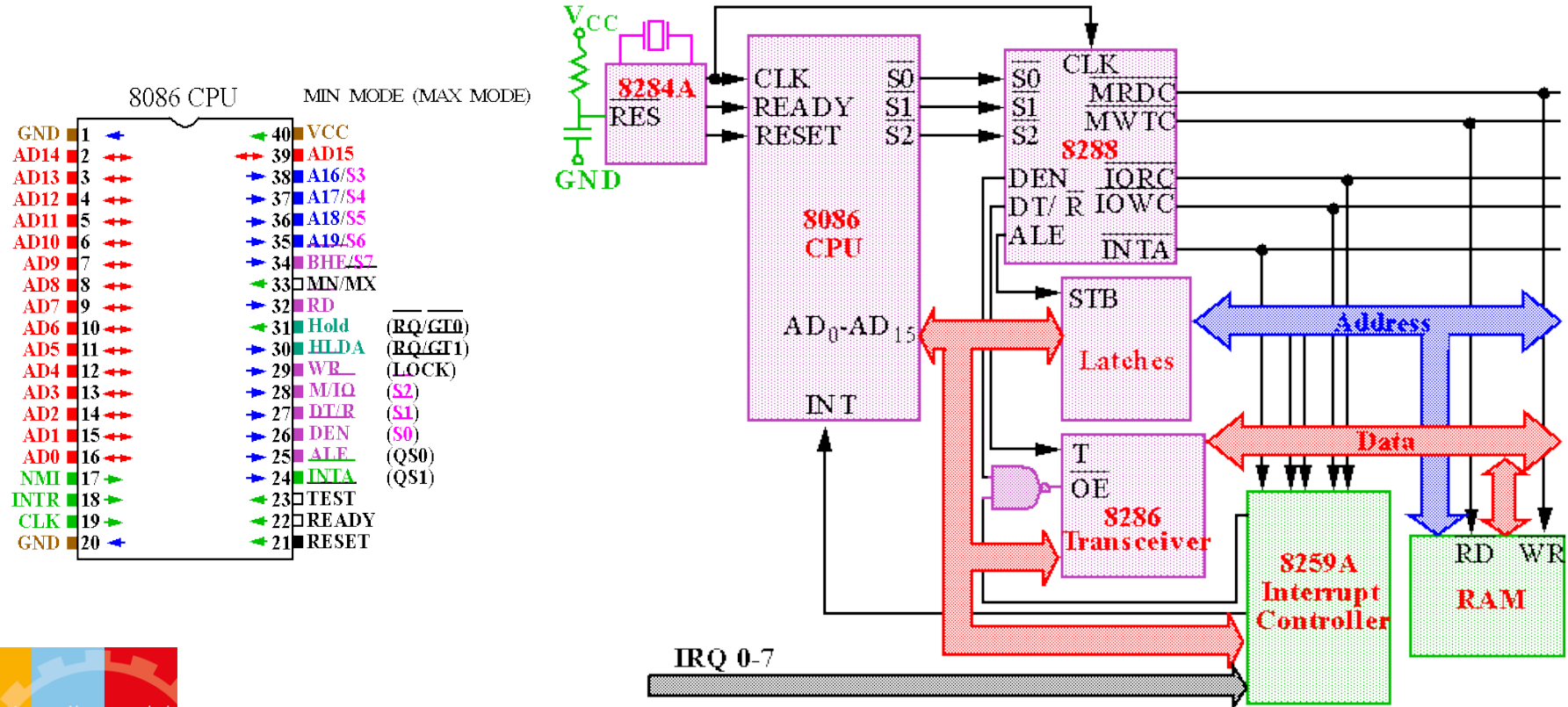
WR: Write control signal, whenever processor **writes data to memory** or I/O port. Active low.

INTA (Interrupt Acknowledge): When the **interrupt request is accepted** by the processor, the output is low on this line

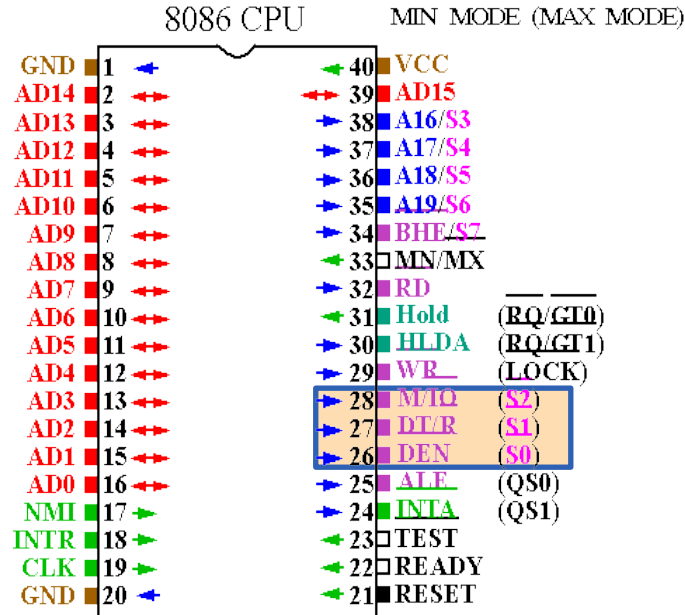
HOLD Input signal to the processor from the bus masters as a request to grant the control of the bus. Usually used by the **DMA controller** to get the control of the bus.

HLDA (Hold Acknowledge) Acknowledge signal by the processor to the bus master requesting the control of **the bus through HOLD**. The acknowledge is asserted high, when the processor accepts HOLD

8086/8088 MAX mode



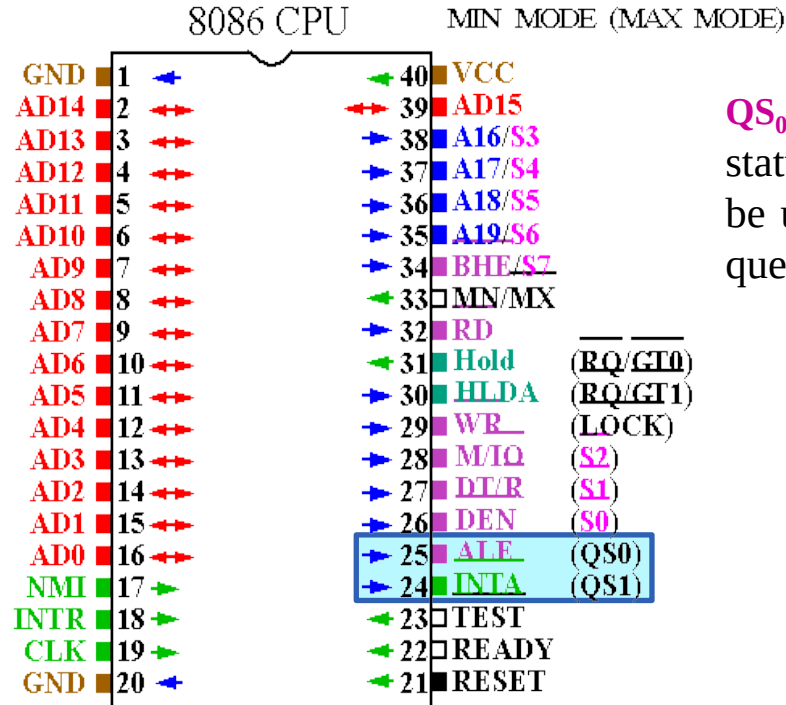
8086/8088 Pin Out (MAX mode)



S_0 , S_1 , S_2 (Status Signals): Active low. Used by the 8086 bus controller to generate bus timing and control signals. These are decoded as shown.

$\overline{S_2}$	$\overline{S_1}$	$\overline{S_0}$	Function
0	0	0	Interrupt acknowledge
0	0	1	I/O read
0	1	0	I/O write
0	1	1	Halt
1	0	0	Opcode fetch
1	0	1	Memory read
1	1	0	Memory write
1	1	1	Passive

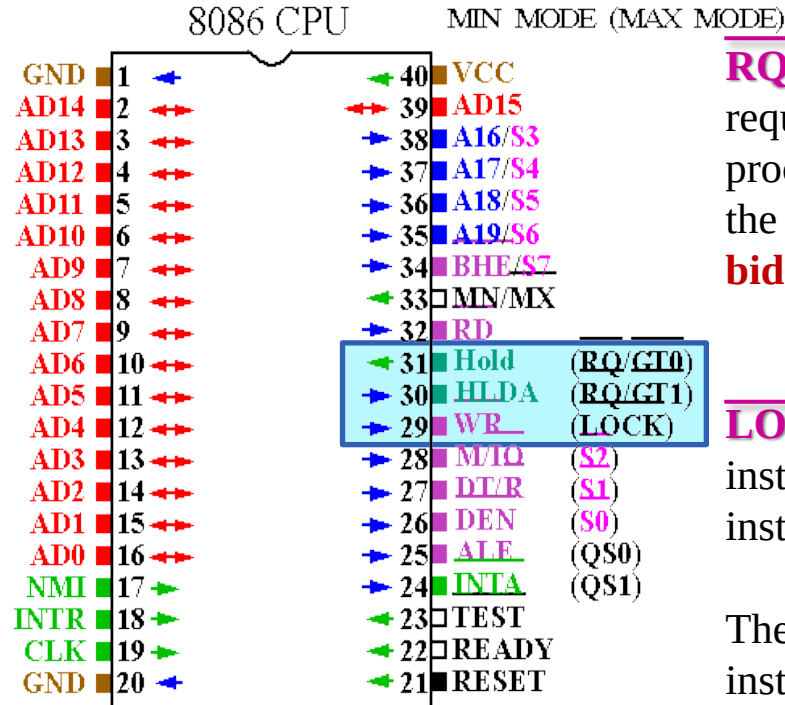
8086/8088 Pin Out (Max mode)



QS₀, QS₁ (Queue Status Signals) The processor provides the status of internal queue in these lines. The queue status can be used by external device to track the internal status of the queue in 8086.

QS ₁	QS ₀	Function
0	0	Queue is idle
0	1	First byte of opcode
1	0	Queue is empty
1	1	Subsequent byte of opcode

8086/8088 Pin Out (MAX mode)



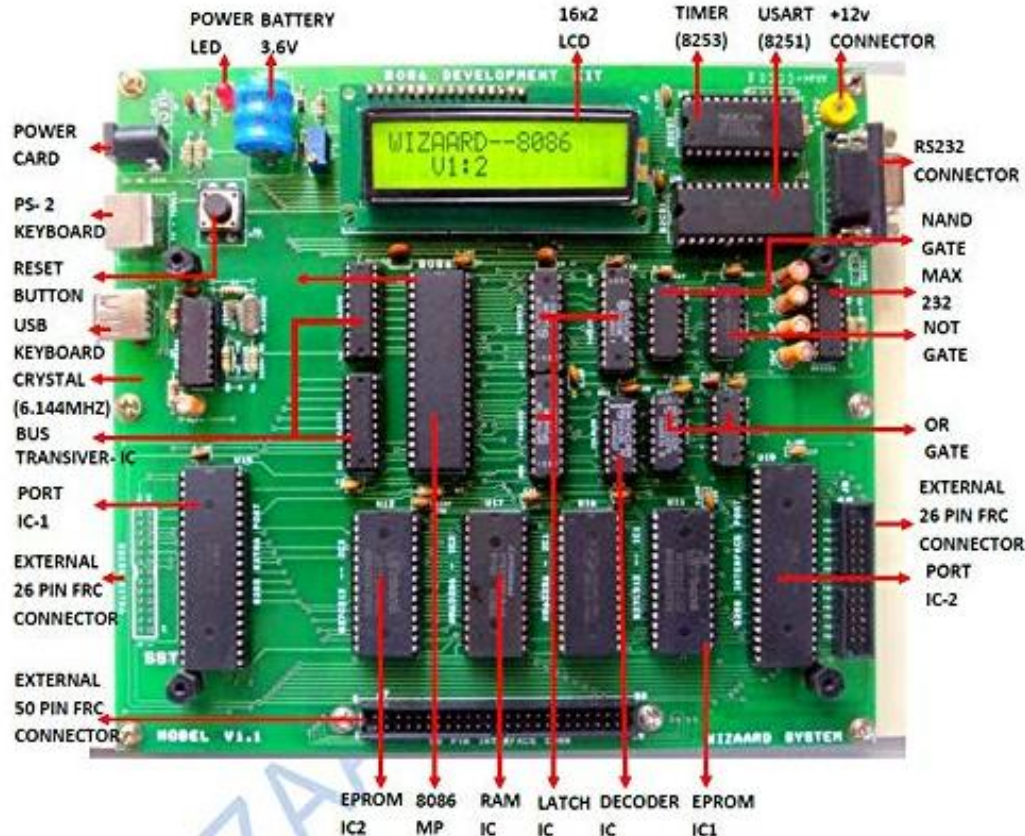
RQ/GT0, RQ/GT1 (Bus Request/ Bus Grant) These requests are used by other local bus masters to force the processor to release the local bus (during DMA) at the end of the processor's current bus cycle. **These pins are bidirectional.**

LOCK: An output signal activated by the LOCK prefix instruction. Remains active until the completion of the instruction prefixed by LOCK.

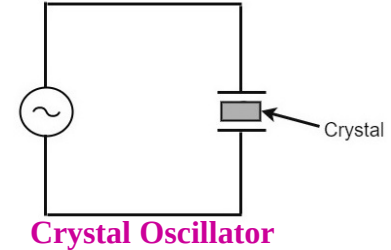
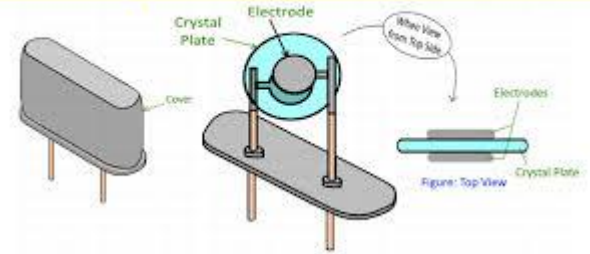
LOCK MOV CX, [4000H];

The 8086 output low on the **LOCK** pin while executing an instruction prefixed by LOCK to prevent other bus masters from gaining control of the system bus.

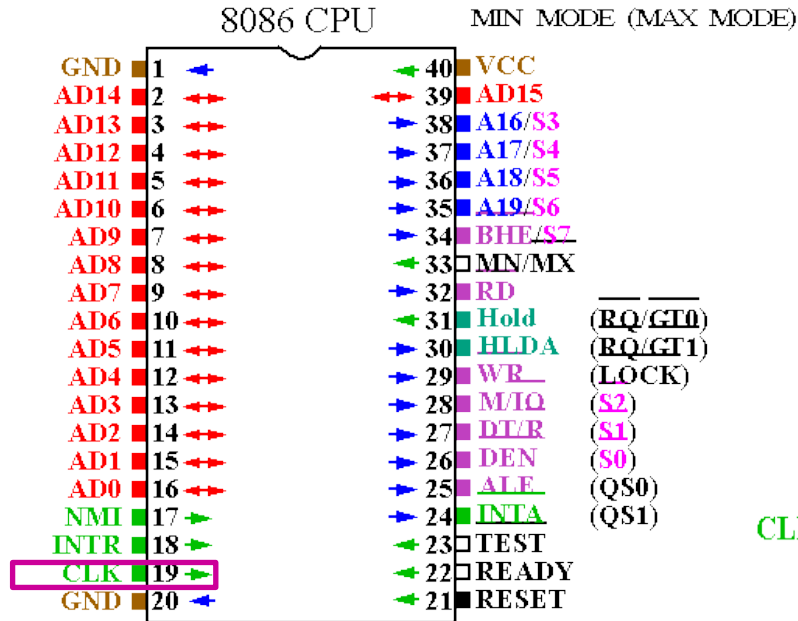
Microprocessor 8086



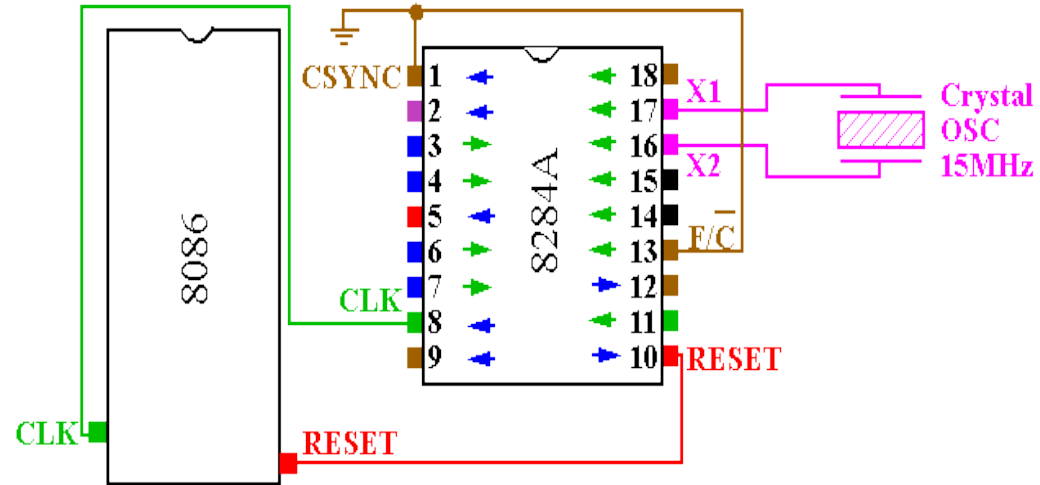
Crystal Oscillator



8284A Clock Generator



Connection of 8284A and the 8086



Basic functions:

- Clock generation.
- RESET synchronization.
- READY synchronization.
- Peripheral clock signal.

Model number	Frequency	Technology
8086	5 MHz ^[13]	HMOS
8086-1	10 MHz	HMOS II
8086-2	8 MHz ^[13]	HMOS II
8086-4	4 MHz ^[13]	HMOS
I8086	5 MHz	HMOS
M8086	5 MHz	HMOS

8284A Clock Generator

Clock generation:

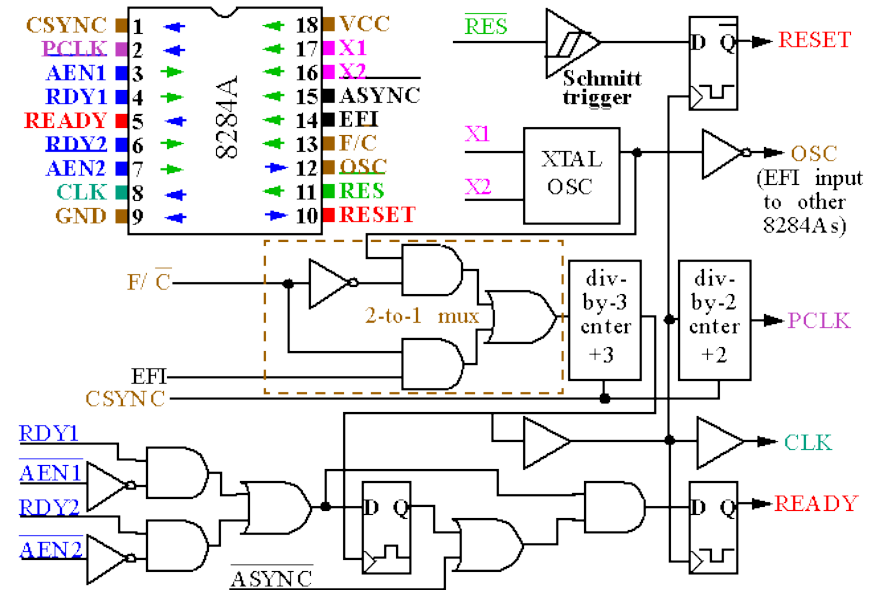
Crystal is connected to **X1** and **X2**.

XTAL OSC generates square wave signal at crystal's frequency which feeds:

- An **inverting buffer** (output **OSC**) which is used to drive the **EFI input** of other 8284As.

- **2-to-1 MUX**

F/C selects XTAL or EFI external input.

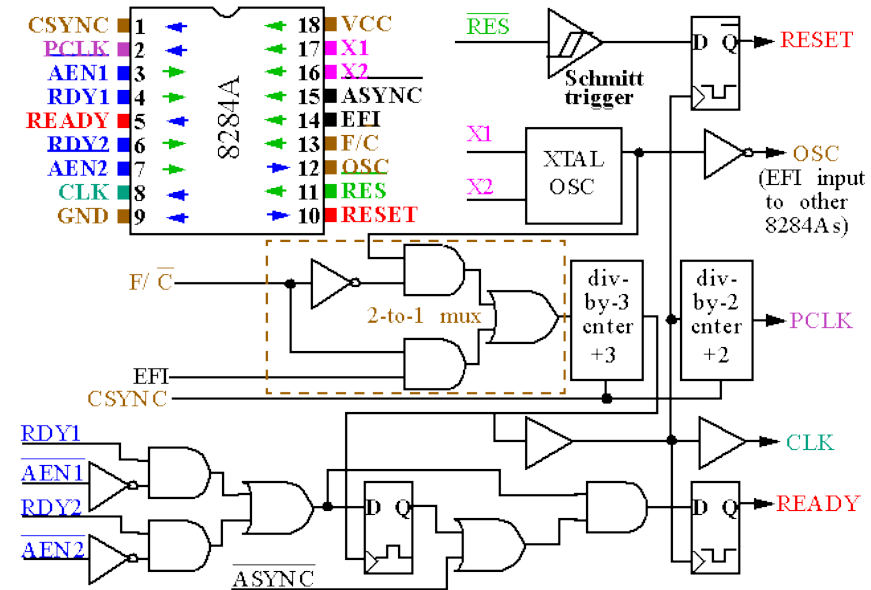


8284A Clock Generator

The MUX drives a divide-by-3 counter (15MHz to 5MHz).

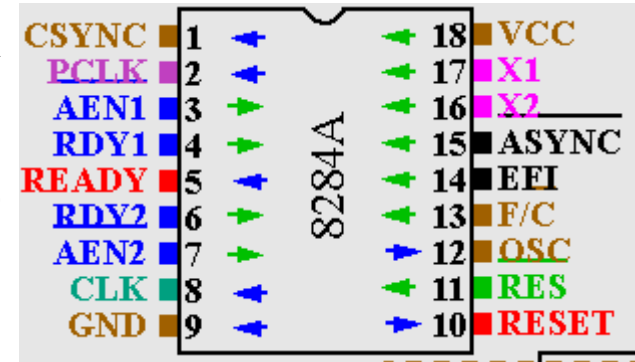
This drives:

- The **READY** flipflop (READY synchronization).
- A second **divide-by-2 counter** (2.5MHz clk for peripheral components).
- The **RESET** flipflop.
- **CLK** which drives the 8086 **CLK** input.



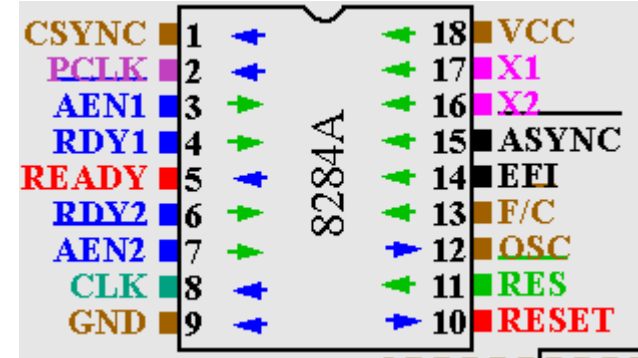
8284A Pin-outs

- $\overline{\text{AEN1}}$ and $\overline{\text{AEN2}}$: The address enable pins provide the bus ready signal RDY1 and RDY2. Wait states are generated by the READY pin which is controlled by these two pins.
- $\overline{\text{ASYNC}}$: This decides one or two stage synchronization. For slow asynchronous device two stage synchronization is used to avoid metastability.
- **PCLK: Peripheral clock signal** is one sixth the crystal or EFI input frequency.
- $\overline{\text{RES}}$ is an active low input to 8284A to provide power on resetting.



Pin-Outs

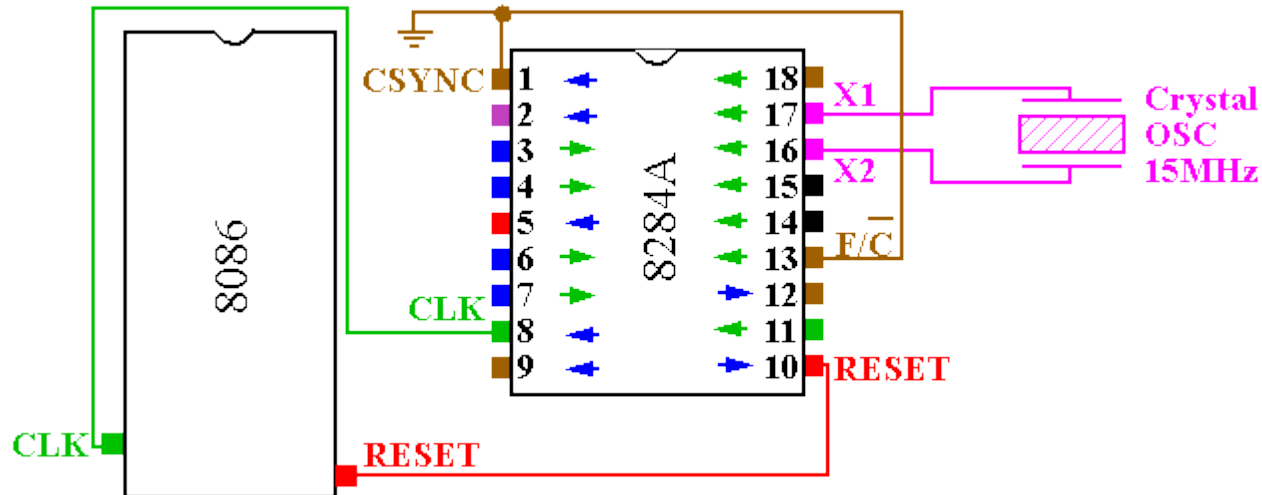
- **CSYNC:** The clock synchronization pin is used whenever the EFI input provides synchronization in systems with multiple processors. This pin should be grounded to select crystal oscillator.
- **OSC:** The oscillator output is a signal that is at the same frequency as crystal. It is used as EFI input to other clock generators in multiple processor system.
- **VCC and GND:** Power supply and ground pin to be connected to +5V and **Ground** respectively.



8284A Clock Generator

RESET:

Negative edge-triggered (1-0) flipflop applies the RESET signal to the 8086 on the falling edge. The 8086 samples the RESET pin on the rising edge (0-1).



Correct reset timing requires that the RESET input to the microprocessor becomes a logic 1 NO LATER than 4 clocks after power up and stay high for at least 50us.